HipMRI: 2D Improved U-Net versus U-Net

Student: Nuoyan Liu (s4688861). **Course:** COMP3710. **Project difficulty:** Normal. **Report updated:** 27 October 2026.

This report compares a six-class 2D Improved U-Net with a standard U-Net baseline. The final test ID5 Dice is **0.844222 versus 0.843161**, an improvement of **0.106 percentage points**, with higher false-positive and computational costs. The evidence supports a cautious prototype comparison rather than a general recommendation to upgrade.

Tutor check-off is completed, as confirmed by the student on 26 October 2026.

The matching PDF is generated from this README and its existing figures.

One-page Written Feasibility Review

Retrospective review, 26 October 2026. This page was assembled from completed experiments with substantial Codex assistance. It is not represented as a contemporaneous midpoint proposal.

User need and scope. An imaging reviewer needs draft pelvic MRI contours for manual inspection. The prototype predicts six-class 2D masks and compares quality with resource cost. It does not make an autonomous clinical-use claim. The primary Normal-level model adapts Isensee et al. (2018) to 2D; the baseline follows the U-Net encoder-decoder idea. Course concepts include convolution, skip connections, residual learning, cross-entropy and validation-based model selection.

Testable acceptance criterion	Evidence and qualification
1. Correct MRI/mask pairing and disjoint case-proxy splits	11,460 training, 660 validation and 540 test pairs audited. True patient identities are not independently verified.
2. Preserve label IDs, use six-channel one-hot CE and exclude padding	Shared implementation and recorded checks; matching Affines have unknown spatial units.
3. Frozen ID5 test Dice at least 0.75	Baseline 0.843161; Improved 0.844222. ID5=prostate remains a disclosed derivative-label assumption.
4. Complete 20 epochs/model, batch 4, FP32 within the available 12 GiB GPU	Completed. Recorded training allocated peaks: 182.642 / 567.992 MiB, not total-device VRAM.
5. Traceable comparison, resources and five diagnostic examples	Per-class Dice/IoU, scoped timing and final epoch-5/17 validation panels are included.

Preliminary evidence and budget. A 64-sample, two-epoch smoke run preceded full runs. The shared full-training budget was 20 epochs per model. The later ID5-selection runs and frozen test, including an authorised Improved-only recovery after a logging failure, are complete. The original pre-training next experiment is documented in the preserved proposal and run plans; this retrospective page does not invent an earlier decision.

Risks and fallback. Label provenance, patient identity proxies, unknown physical units, one seed, three test groups and prior test exposure limit conclusions. Improved gains only 0.106 percentage points in primary Dice while increasing false positives and resource use. Keep the baseline as a low-cost reference and send difficult contours for human review. The current next step is submission/document review, with no further training or test evaluation proposed. New generalisation experiments would require a separate design and independent data.

1. Data, labels and preprocessing

The input is a single-channel 2D NIfTI MRI slice; the target is one mutually exclusive raw class ID per pixel. Filename pairing replaces the `case_` prefix with `seg_` while retaining exactly the same case, week and slice identifiers. Equal array shapes alone do not establish a pair.

Split	Paired slices	Numeric case-proxy groups	Use
Training	11,460	31	Parameter updates
Validation	660	4: cases 36-39	Checkpoint selection and diagnostics
Test	540	3: cases 40-42	Frozen final comparison

The case sets are pairwise disjoint. Different weeks/slices of one case must not be treated as independent patients. The filename case number is an identity proxy; no independent patient crosswalk is available. The test manifest audit records 540 pairs of shape 256 by 128, containing 17,694,720 original pixels. These pixels are not independent subjects.

The provider's original label schema is:

ID	Provider class	ID	Provider class
0	Background	3	Bladder
1	Body	4	Rectum
2	Bones	5	Prostate

The label schema and source record document the CSIRO mapping. Applying it to the course's derived `keras_slices_data` assumes that export preserved the original IDs. This assumption was accepted for the project but is not independently proven. Masks were not relabelled. Reported primary scores therefore refer precisely to **ID5**, interpreted as prostate under that assumption.

`dataset.py` checks matching shapes and finite 4-by-4 Affines before pixel loading, using absolute tolerance 0.00001 and relative tolerance zero. The training/validation geometry audit found matching identity Affines. Exported spatial units are `unknown`: header agreement is not independent proof of anatomical registration, orientation or distances in millimetres.

Preprocessing is shared by both models:

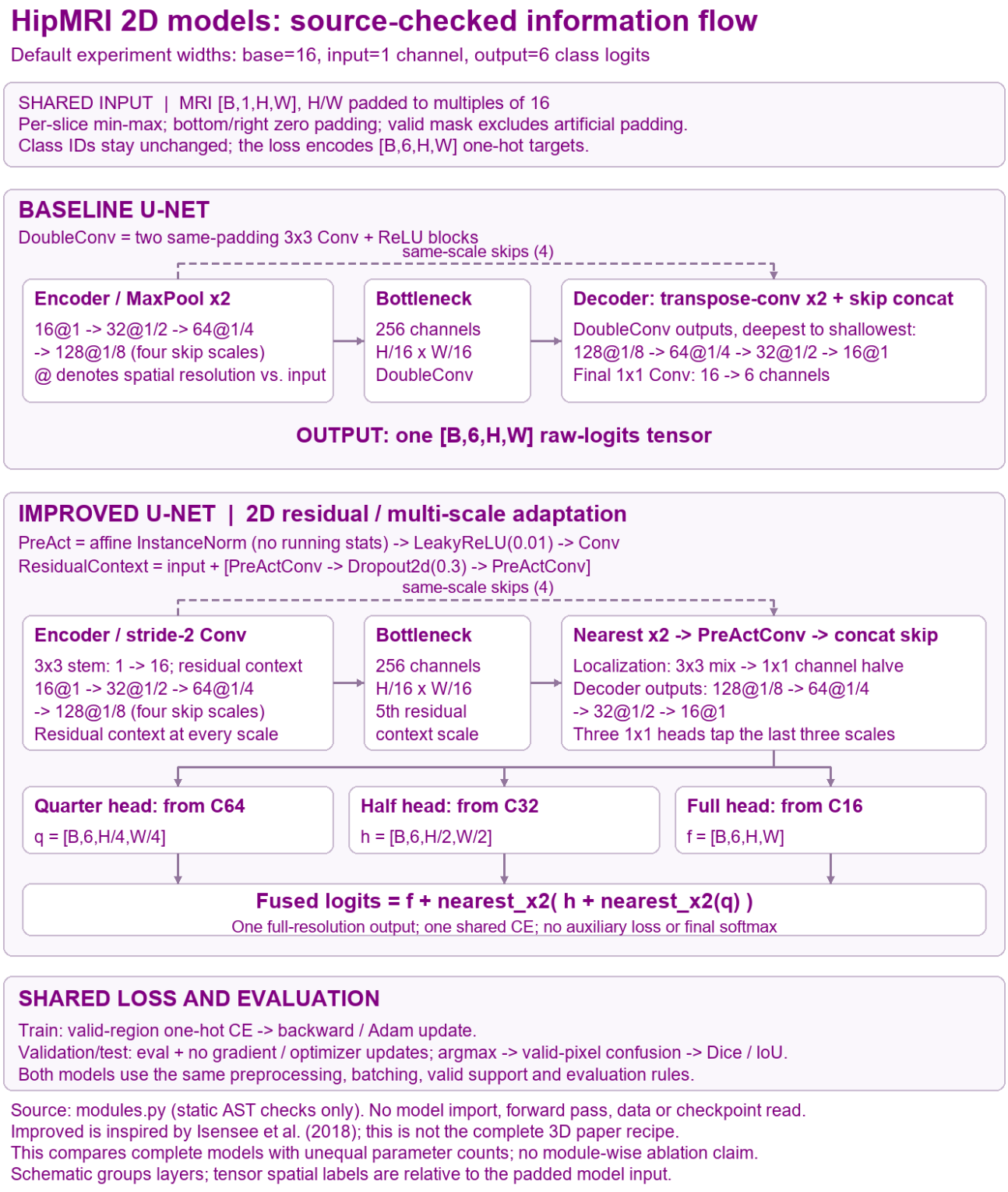
1. Load paired arrays and preserve raw integer mask IDs.
2. Apply per-slice MRI min-max normalisation: `(x - minimum) / (maximum - minimum)`. Constant slices become zero. Masks are not intensity-normalised.
3. Pad only the bottom/right to multiples of 16; collate pads samples further to the largest shape in the batch. Most slices are 256 by 128; 60 training slices are 256 by 144. These sizes already satisfy the model's factor-16 requirement, but mixed batches still need a common shape.
4. Set added MRI pixels to zero, target pixels to -100 and their valid mask to false. Every original pixel is valid, including background and zero-intensity pixels. Preserve original dimensions for cropping predictions back without interpolation.

The dataset retains integer targets. `losses.masked_one_hot` constructs `[B,6,H,W]` targets during the loss call; padding has six zeros. Model inputs are `[B,1,H,W]`, and outputs are six raw logits per pixel. Cross-entropy is averaged over valid pixels only. The shared implementation is value/gradient equivalent to the corresponding ignore-index CE, as recorded in earlier checks. No augmentation is used in the frozen protocol.

2. Models and shared learning procedure

The baseline uses four MaxPool downsampling stages, double 3-by-3 convolution/ReLU blocks, transposed-convolution upsampling and same-scale encoder feature concatenation. It uses same padding and base width 16; it is not the complete original U-Net training recipe.

Improved U-Net is a **2D adaptation** of the residual, multi-scale architecture described by Isensee et al. (2018): five context scales; pre-activation InstanceNorm with affine parameters and no running statistics; LeakyReLU slope 0.01; residual context blocks; Dropout2d 0.3; stride-2 convolution downsampling; nearest-neighbour upsampling followed by convolution; encoder-decoder concatenation; and localisation blocks using 3-by-3 then 1-by-1 convolutions with channel reduction. Quarter-, half- and full-resolution classification heads are progressively upsampled and summed as logits. One shared CE supervises the final fused logits; there are no separate auxiliary losses.



Source-based diagram of the two implementations and their shared pipeline

The editable architecture diagram follows modules.py. Both models use base width 16, one input channel and six classes, but have different parameter counts. This compares two complete implementations, not a capacity-matched ablation or a causal test of one component. The Improved model is not a full reproduction of the original 3D BraTS loss/augmentation/training recipe.

Shared condition	Fixed setting
Optimisation	Valid-region one-hot cross-entropy; Adam; learning rate 0.001; weight decay 0; seed 3710
Budget	20 complete epochs/model; batch 4; 2,865 updates/epoch; 57,300 total updates/model
Other settings	FP32; workers 0; no augmentation, AMP, scheduler or early stopping
Validation/test	eval mode, gradients disabled, no optimiser updates; lexicographic order, batch 4, no shuffle, keep last batch

Shared condition	Fixed setting
Prediction	Six-class argmax; original valid region; no TTA or threshold tuning

The recorded environment is Python 3.13.2, NumPy 2.5.4, NiBabel 5.4.2, PyTorch 2.11.0+cu128, CUDA 12.8, cuDNN 91900 and an RTX 4070 Ti with 12 GiB. CPU threads were 1; cuDNN deterministic was true and benchmark false. Matmul TF32 was false, cuDNN TF32 true and float32 matmul precision `highest`. FP32 refers to tensor dtypes with AMP disabled, not to disabling every backend TF32 setting. Exact observations are in the run record and preflight.

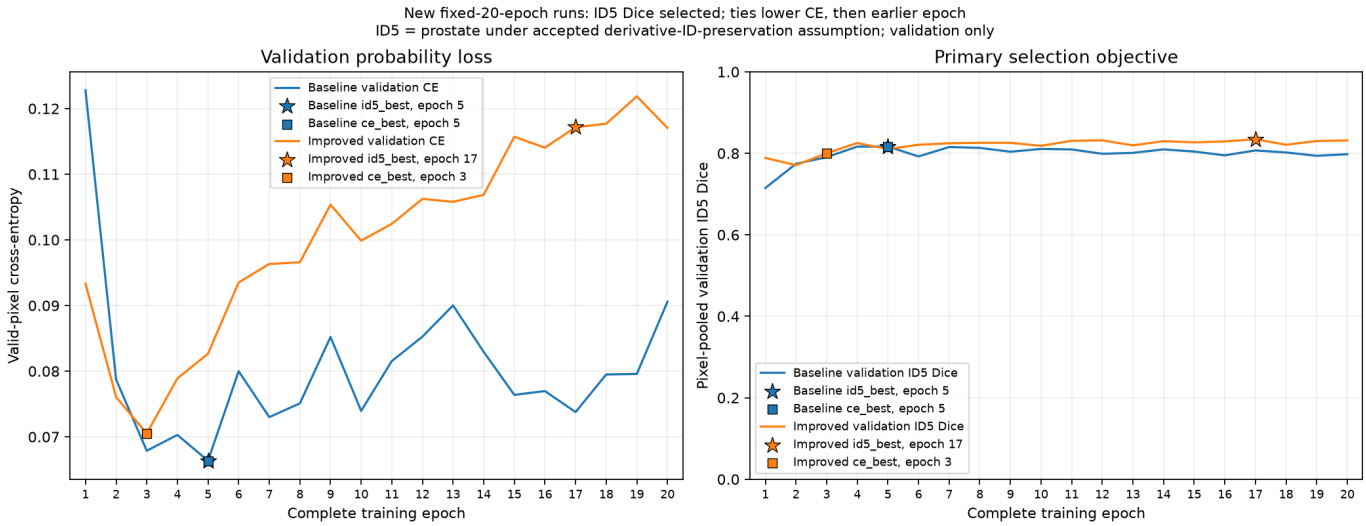
For each split, accumulate a confusion matrix across all valid pixels before calculating class Dice and IoU. For one class, `Dice = 2TP / (2TP + FP + FN)` and `IoU = TP / (TP + FP + FN)`. Foreground macro scores exclude ID0 and undefined classes with both prediction/reference empty. A class with an empty reference but positive predictions scores zero. These are pooled-pixel scores, not averages of batch Dice or equally weighted patient scores. CE is weighted by valid-pixel counts and cannot be reconstructed from hard predictions. Improved's InstanceNorm sees padded inputs, so batch composition and padding are part of the shared protocol.

3. Validation selection and freezing

The original full experiment selected minimum validation CE, yielding baseline epoch 5 and Improved epoch 3. Diagnostics showed that Improved's best ID5 overlap occurred later than its best CE. Before the final test, the student authorised another 20-epoch run for each model under the same settings with a common selection rule: **maximum validation ID5 Dice; then minimum CE for an exact tie; then earliest epoch**. Comparisons use unrounded values. ID5-best, CE-best and last checkpoints are saved separately; `best.pt` is the selected-metric alias.

Selection stage	Baseline epoch / validation ID5 Dice	Improved epoch / validation ID5 Dice
Original minimum-CE experiment	5 / 0.816567	3 / 0.800187
Final common ID5-selection experiment	5 / 0.816567	17 / 0.834394
Preserved checkpoint roles	ID5-best 5; CE-best 5; last 20	ID5-best 17; CE-best 3; last 20

The final validation ID5 advantage is 1.783 percentage points, while foreground macro Dice is 0.903274 versus 0.896437, favouring the baseline. The original comparison and ID5 comparison are both preserved. The common-history metrics and CE-best/last model and Adam states matched their earlier runs: this selection change is not a new training-algorithm gain, and repeating one seed is not an independent-seed replication.



Twenty-epoch training/validation curves and the final ID5 selection

The frozen test recipe binds baseline epoch 5 and Improved epoch 17, checkpoint hashes, preprocessing and scoring rules. Epochs 5/17 were selected from 20 completed epochs; training did not stop early at those epochs. Test results did not choose another epoch, architecture, threshold or learning rate.

4. Frozen test results

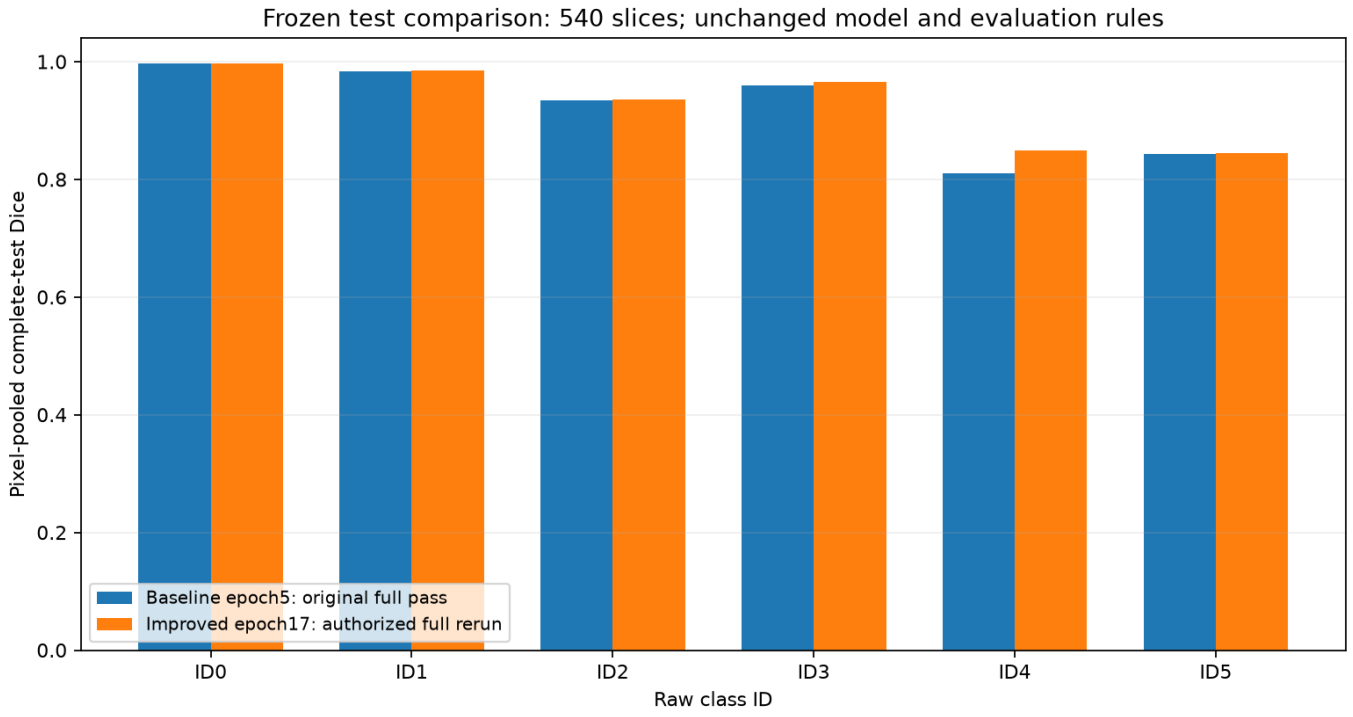
Complete test metric	Baseline epoch 5	Improved epoch 17	Improved minus baseline
ID5 Dice	0.843161	0.844222	+0.001060 = +0.106 percentage points
ID5 IoU	0.728850	0.730435	+0.001586
Foreground macro Dice	0.906227	0.915912	+0.009685
Foreground macro IoU	0.835422	0.850269	+0.014847
Valid-pixel CE, lower is better	0.055552	0.071634	+0.016081, worse
Pixel accuracy	0.977617	0.978941	+0.001325

Raw class ID	Baseline Dice / IoU	Improved Dice / IoU
0	0.996669 / 0.993360	0.997206 / 0.994427
1	0.983693 / 0.967909	0.984636 / 0.969737
2	0.934470 / 0.877000	0.936415 / 0.880433
3	0.959706 / 0.922534	0.965891 / 0.934031
4	0.810103 / 0.680817	0.848396 / 0.736707
5	0.843161 / 0.728850	0.844222 / 0.730435

All pooled class Dice scores are higher for Improved, with ID4 gaining 3.829 percentage points. This does not imply that every slice or case improves. Primary ID5 Dice increases by approximately 0.1257% relative to baseline, which is distinct from the absolute gain of 0.106 percentage points. CE is approximately 28.95% higher; a loss difference is not expressed in percentage points.

ID5 statistics from the saved test confusion matrices	Baseline	Improved
TP / FP / FN	48,408 / 6,382 / 11,627	53,099 / 12,660 / 6,936
Precision	0.883519	0.807479
Recall	0.806330	0.884467

Both references contain 60,035 ID5 pixels. Improved recovers 4,691 additional true pixels but adds 6,278 false positives, explaining the small net Dice gain. These are pixel-level descriptions, not patient diagnostic sensitivity or specificity. Each complete scoring pass covers 540 slices in 135 batches with zero updates and unchanged model state. The complete comparison retains full precision and source bindings.



Per-class pooled test Dice from the completed baseline pass and authorised Improved recovery

Test interruption, recovery and exposure

The baseline completed its original 540-slice pass. Improved produced 444 outputs over 111 batch forwards before a Windows `PermissionError` during atomic logging replacement terminated that attempt. The 111th batch had not returned to the shared scoring loop; earlier aggregate statistics were lost with the process. A valid full-split score could not be reconstructed from rounded console output or the remaining samples. The original attempt remains incomplete.

The failure arose in Codex-added persistence code. The fix used unique same-directory temporary files, flush/fsync and bounded replacement retries; model forwards were not retried by that I/O fix. The specific external cause of file denial was not established. After the repeated exposure was explained, the student explicitly authorised **Improved-only full recovery** with the same frozen epoch-17 weights and rules. It completed 540 slices; baseline was not rerun. Improved therefore had 246 batch forwards and 984 sample exposures across both attempts, including 444 repeats. Final Improved scores use only the complete authorised recovery, not reconstructed partial scores. It would be inaccurate to say both models were tested exactly once.

The failure record, additional authorisation and recovery record remain available. No test-driven tuning followed.

5. Resource cost and timing scope

Measurement	Baseline	Improved	Ratio
Parameters	1,940,902	3,190,194	1.644
Training peak allocated MiB	182.642	567.992	3.110
Training peak reserved MiB	250.000	694.000	2.776
Validation peak allocated MiB	92.182	141.250	1.532
Restore-phase allocated MiB	52.998	84.844	1.601
20-epoch run wall-clock seconds	2,243.088	3,549.024	1.582

These values come from 42 synchronised phase records/model in the ID5 comparison. Phase scope includes the model, Adam/gradient/activation and allocator state specified in each record. Validation includes resident training state; restoration includes its additional model. Reserved memory includes cache. The measurements are not total-device VRAM, incremental memory or an unmeasured whole-process peak. Both models completed the available GPU budget. Wall-clock ratios are run observations rather than pure compute-speed measurements.

Synthetic input	Baseline median / p95 batch ms	Improved median / p95 batch ms	Median ratio
B1, 256 by 128	1.146 / 3.072	4.776 / 8.876	4.169
B1, 256 by 144	1.432 / 1.574	11.024 / 12.495	7.697
B4, 256 by 128	1.347 / 1.477	10.747 / 12.402	7.977
B4, 256 by 144	3.134 / 4.028	10.756 / 12.267	3.432

The baseline profile and Improved profile bind the frozen checkpoints. Both use the same RTX 4070 Ti/runtime, FP32 eval inference, 10 warmups and 50 synchronised timed forwards per shape; p95 uses linear interpolation. They exclude disk I/O, weight loading, transfers, metrics and plotting, and do not read test pixels. Raw timings were independently re-aggregated. B1 represents a synthetic single-slice forward; B4 is an entire batch. Neither is full-patient/sequence end-to-end latency. This single local snapshot does not isolate all system load or provide repeat-run confidence intervals. Ratios of 3.43-7.98 do not generalise automatically to other hardware.

Complete-test elapsed times of 3.611 and 8.732 seconds are not used as a fair latency comparison because the recovered Improved run used different persistence/retry logging and runtime/cache conditions.

6. Five final-model validation examples

The five A-E cases and their order were fixed from the earlier diagnostic list **before** generating the final epoch-5/17 displays. The display plan and execution record describe this later validation-only supplement. Each frozen model made five B1 forwards, ten total, in eval/inference mode, with no optimiser, backward pass, gradients or state changes. The ten unique validation MRI/mask files were loaded; no real test images were read. Independent GT and predictions come from original-size arrays, not reconstructed PNGs.

Case / week / slice	GT ID5 pixels	Baseline TP / FP / FN	Improved TP / FP / FN	Baseline Dice	Improved Dice
A: 038 / 0 / 29	160	0 / 0 / 160	58 / 0 / 102	0.000000	0.532110
B: 039 / 5 / 41	0	0 / 269 / 0	0 / 175 / 0	0.000000	0.000000
C: 037 / 0 / 9	19	0 / 0 / 19	0 / 0 / 19	0.000000	0.000000
D: 039 / 7 / 25	184	103 / 8 / 81	153 / 9 / 31	0.698305	0.884393
E: 037 / 0 / 17	342	241 / 108 / 101	282 / 220 / 60	0.697540	0.668246

Each original six-column figure shows the full slice above a shared display ROI. Columns show MRI, independent six-class GT, baseline prediction and ID5 errors, then Improved prediction and ID5 errors. Green is TP, magenta FN and cyan FP; these are error colours, not new class IDs. The ROI is the reference/prediction ID5 union plus an eight-pixel margin; all metrics use the full original valid region. These are qualitative contour/error comparisons, not an independent distance-based boundary metric. Physical units and base/apex anatomy remain unverified.

Added offline contour diagnostics

This analysis was added on 27 October 2026 using only the five original-size NPZ files saved on 26 October 2026. Each compressed-file SHA256 matches the earlier execution record; the original six-class confusion matrices and region Dice values were independently rechecked from those same arrays. This work read zero raw NIfTI files and zero checkpoints, imported no model, and performed zero model forwards, optimiser updates, training calls or test-set reads/evaluations. The ten validation forwards described above belong to the earlier 26 October execution, not this offline analysis. The new figures and numbers are bound in the offline diagnostic record.

The following custom pixel-grid contour diagnostic is defined for raw ID5. A boundary pixel is an ID5 pixel with at least one of its four edge-neighbours outside ID5; the exterior of the image is background. Compute both boundaries over all original pixels, never the display crop or artificial padding. Precision is the fraction of predicted boundary pixels with at least one reference boundary pixel in the matching window; recall reverses prediction and reference. F1 is their harmonic mean, with the empty-set conventions below. Exact matching uses the same pixel location. The 1px result uses Chebyshev distance at most one pixel, a 3-by-3 square including diagonals, not a Euclidean radius of one pixel and not a millimetre tolerance. Matching is not one-to-one: several boundary pixels may match one nearby counterpart.

The added detail figures draw the binary masks as 0.5-level contours: white is the reference ID5 contour, cyan the baseline contour and orange the Improved contour; red and magenta show FP and FN regions. Those interpolated drawing lines are a different representation from the four-neighbour inner-boundary pixels counted in the table; no metric is measured from the rendered lines. Coordinates are zero-based original array rows and columns with inclusive bounds. They do not establish patient left/right, superior/inferior, base or apex. Organ names below follow the existing provider-ID preservation working assumption. Region class-confusion counts describe where labels went; they are distinct from the actual boundary-pixel

matching scores.

NA means undefined. With a nonempty reference and empty prediction, boundary precision is NA, recall is zero and F1 is set to zero. With an empty reference and nonempty prediction, precision is zero while recall and F1 are NA; case B therefore has no reference contour to assess. Both empty sets would give NA for all three. Each row is a post-hoc diagnostic of one of the same five selected validation examples. No mean over these cases, full-validation score or test score is calculated, and no checkpoint, threshold or model was reselected.

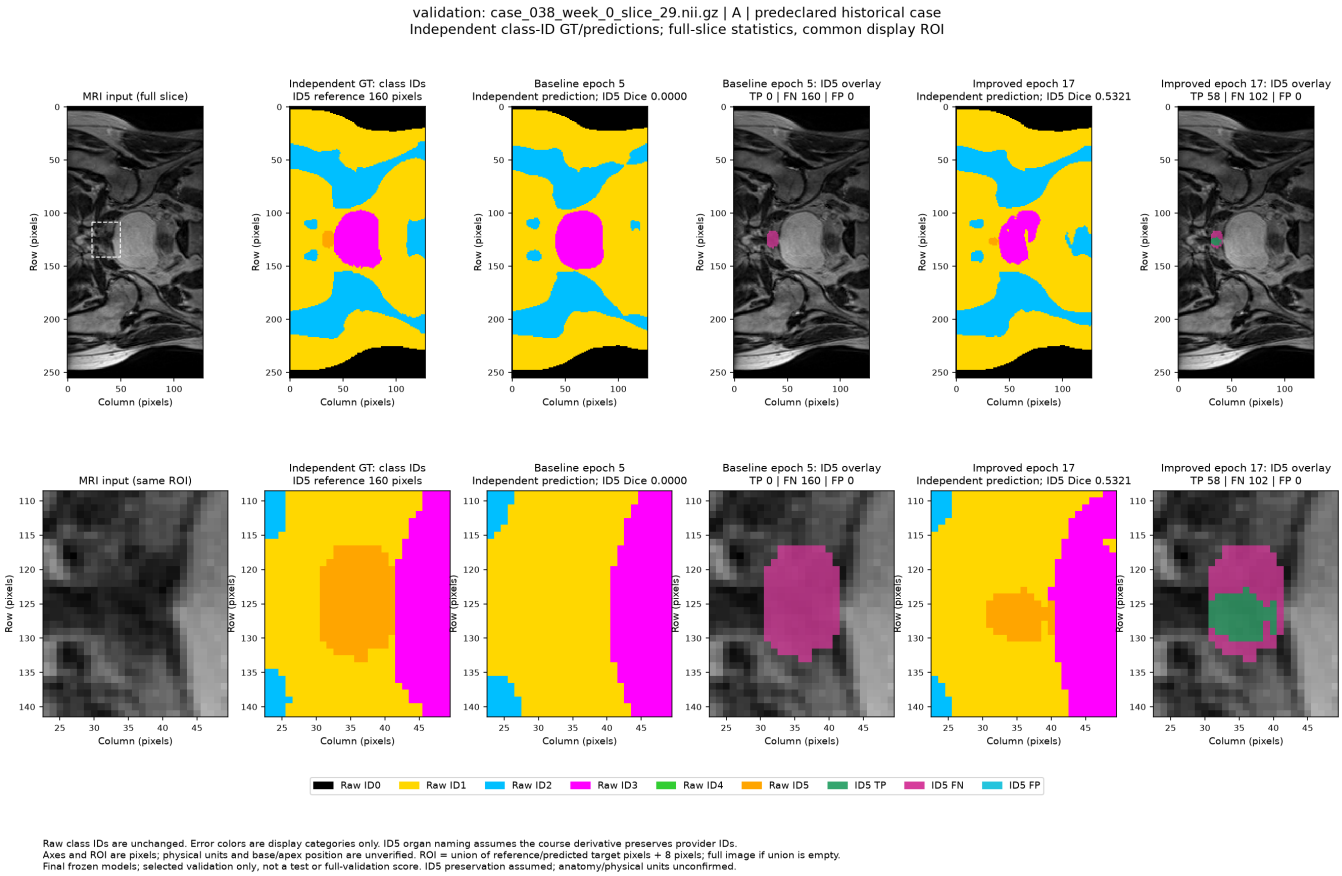
Case / model	Reference / predicted boundary pixels	Exact P / R / F1	1px P / R / F1
A / Baseline 5	43 / 0	NA / 0.000 / 0.000	NA / 0.000 / 0.000
A / Improved 17	43 / 27	0.148 / 0.093 / 0.114	0.481 / 0.395 / 0.434
B / Baseline 5	0 / 53	0.000 / NA / NA	0.000 / NA / NA
B / Improved 17	0 / 44	0.000 / NA / NA	0.000 / NA / NA
C / Baseline 5	15 / 0	NA / 0.000 / 0.000	NA / 0.000 / 0.000
C / Improved 17	15 / 0	NA / 0.000 / 0.000	NA / 0.000 / 0.000
D / Baseline 5	47 / 36	0.333 / 0.255 / 0.289	0.556 / 0.426 / 0.482
D / Improved 17	47 / 40	0.450 / 0.383 / 0.414	0.825 / 0.830 / 0.827
E / Baseline 5	63 / 65	0.123 / 0.127 / 0.125	0.292 / 0.397 / 0.337
E / Improved 17	63 / 74	0.081 / 0.095 / 0.088	0.311 / 0.429 / 0.360

The one-pixel tolerance is particularly forgiving for tiny regions: as a constructed arithmetic counterexample, a 3-by-3 reference square and a single central predicted pixel have contour F1 at 1px equal to 1, yet region Dice is only 0.2. This is a numeric warning, not an MRI result. Retain both exact/tolerant contour scores and the unchanged region metrics rather than selecting the most favourable number.

A. Missed reference region: case 038, week 0, slice 29

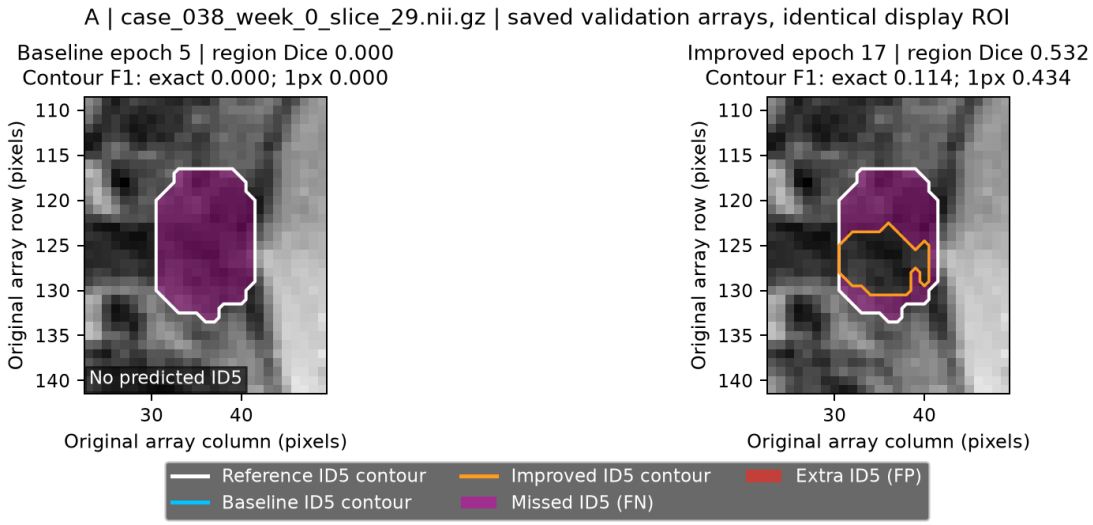
The reference ID5 region spans rows 117-133 and columns 31-41. Baseline predicts no ID5 contour. Improved recovers 58 pixels entirely within the reference, but its extent is only rows 123-130 and columns 31-40. The smaller- and larger-row ends remain truncated, with a small local missed patch/indentation near the recovered contour. It has no FP, yet misses 102 of the 160 reference pixels.

The reference ID5 boundary adjoins IDs 1 and 3. Improved assigns 90 missed pixels to ID1 (Body) and 12 to ID3 (Bladder). Its exact contour P/R/F1 are 0.148/0.093/0.114; with the defined 1px window they are 0.481/0.395/0.434. The wider-window matches do not recover the missing target region. A reviewer should redraw the truncated extent against the reference and inspect the local ID5/ID3 interface before accepting a draft; this example does not establish base/apex anatomy.



Validation A: complete MRI, GT, baseline-5 and Improved-17 predictions and error panels

Added boundary detail for case A, generated from the unchanged saved validation arrays; no new inference. The two models use the same display ROI, while all numbers use the full original image.

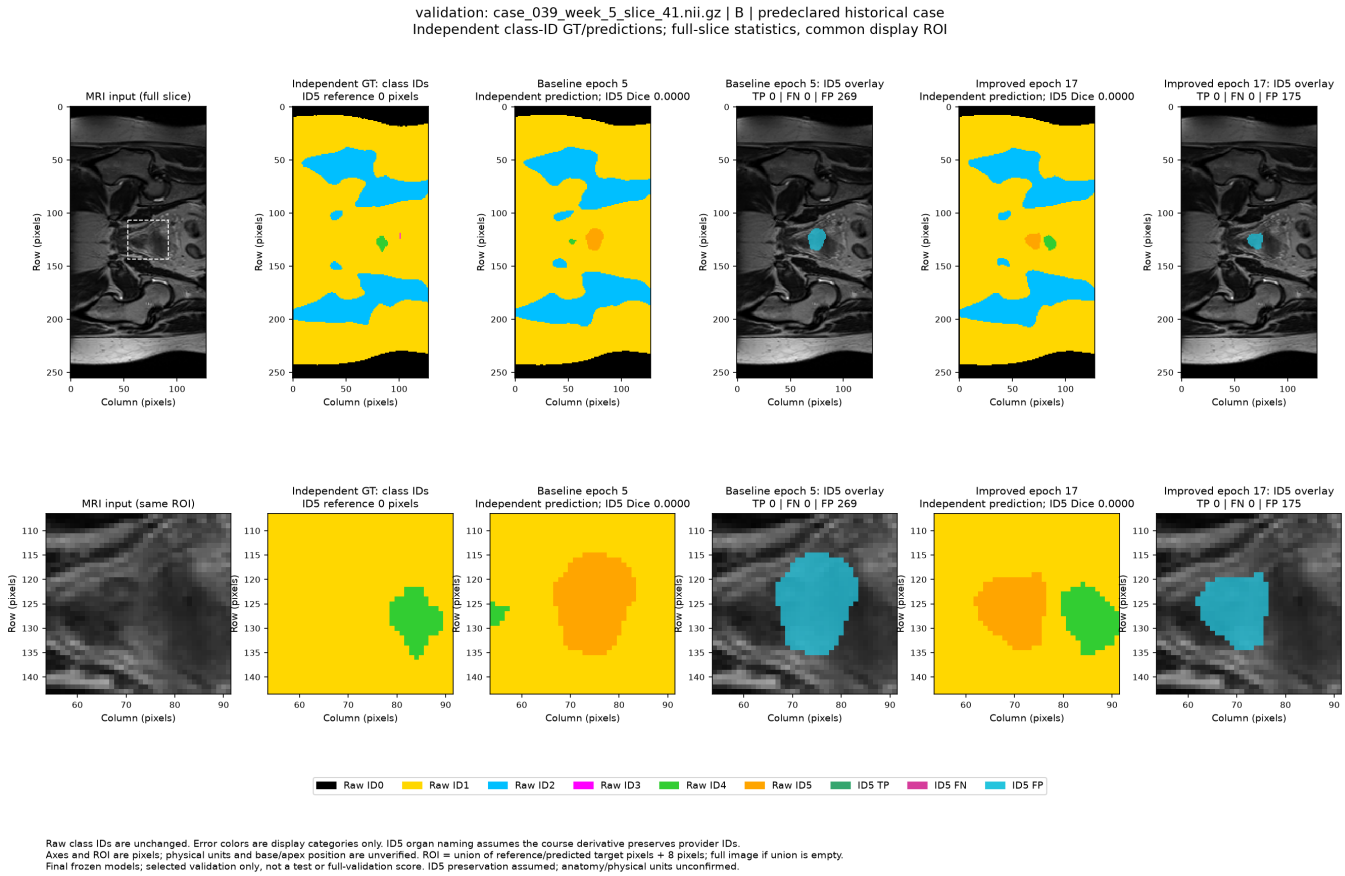


Added offline boundary detail for validation case A: reference and frozen-model contours from the same saved arrays

B. False positives with empty reference: case 039, week 5, slice 41

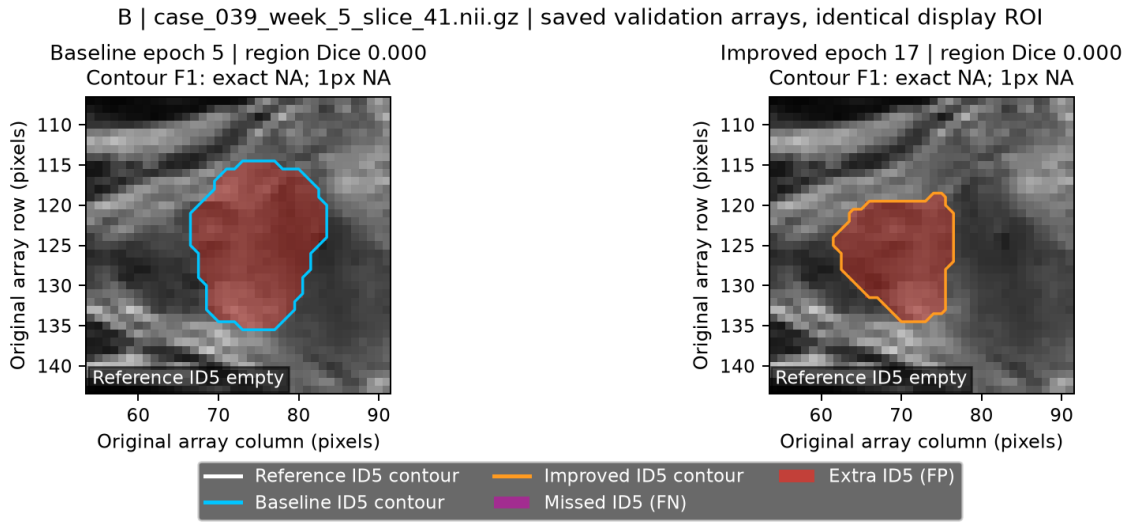
This reference-empty slice has no ID5 contour. Baseline nevertheless produces a connected 269-pixel ID5 region at rows 115-135, columns 67-83: 245 FP pixels overwrite reference ID1 (Body) and 24 overwrite reference ID4 (Rectum). Improved produces a connected 175-pixel region at rows 119-134, columns 62-76, entirely over reference ID1. Fewer FP and the absence of ID4 overlap in Improved do not make its nonempty draft correct.

Boundary recall and F1 are NA because there is no reference contour to match; boundary precision is zero for both. This is a spurious-region failure rather than a measured displacement of an existing reference contour. Automatic acceptance should be withheld while a reviewer checks the false draft against the annotation and local anatomy. An empty annotation does not by itself prove anatomical absence, and the array-coordinate shift is not a patient-direction claim.



Validation B: empty ID5 reference and residual false positives

Added boundary detail for case B, generated from the unchanged saved validation arrays; no new inference. The two models use the same display ROI, while all numbers use the full original image.

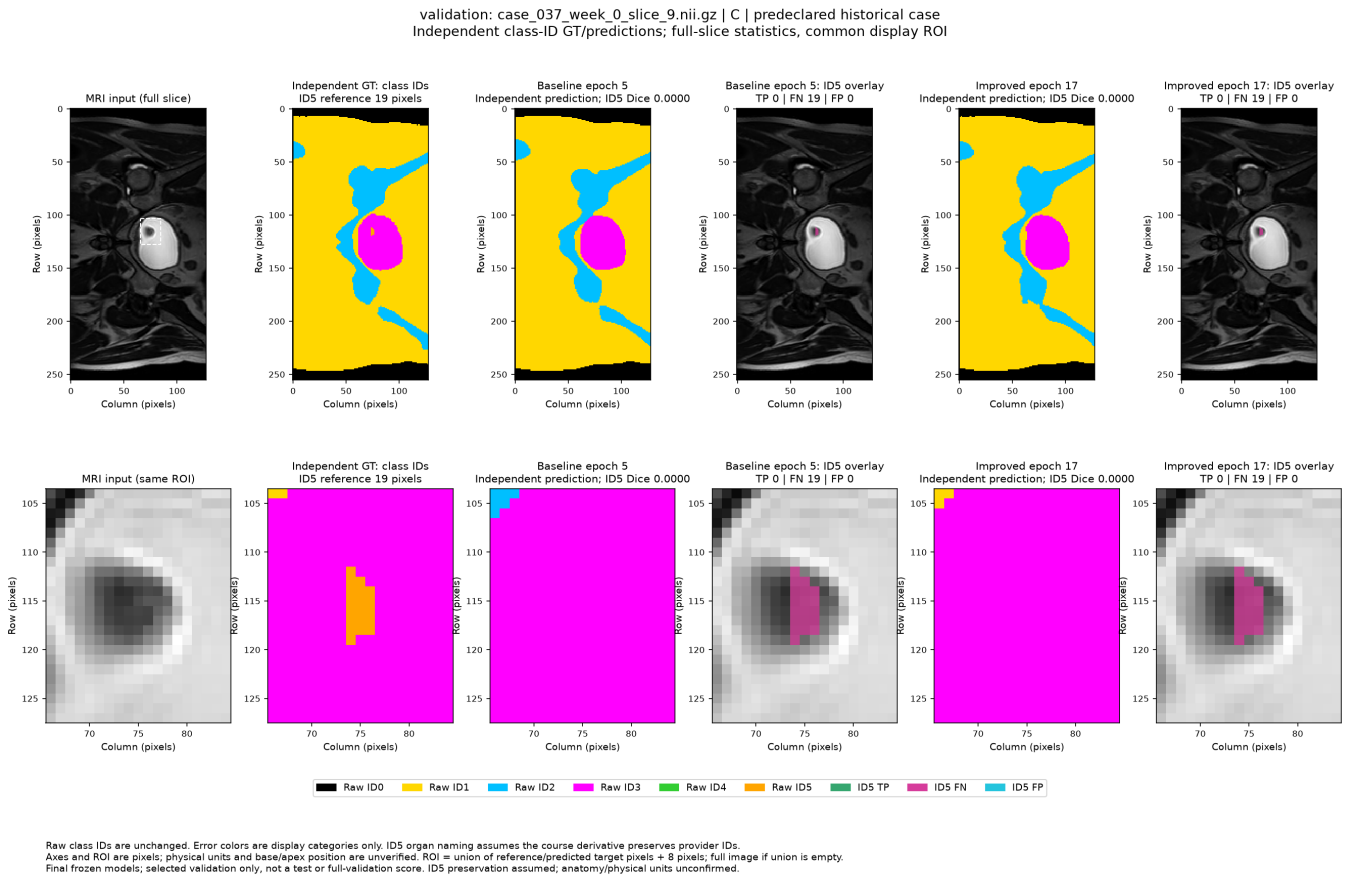


Added offline boundary detail for validation case B: reference and frozen-model contours from the same saved arrays

C. Very small reference region: case 037, week 0, slice 9

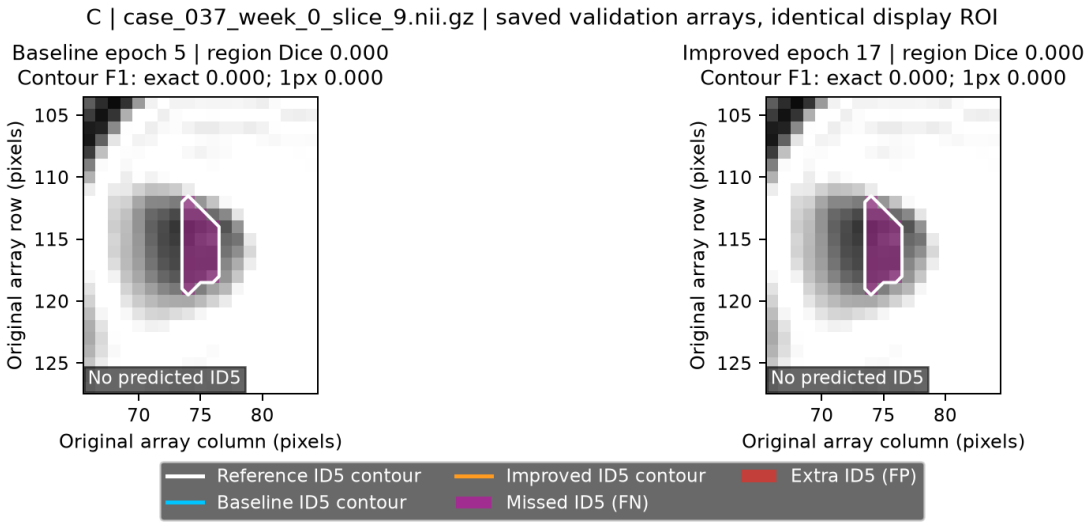
The 19-pixel reference ID5 region lies at rows 112-119, columns 74-76, with 15 four-neighbour inner-boundary pixels. Its annotated boundary adjoins ID3 (Bladder under the working mapping). Both models assign all 19 reference pixels to ID3 and predict no ID5 region, so the reference ID5/ID3 interface disappears entirely from the draft. Exact and 1px boundary recall/F1 are zero; precision is NA because no predicted boundary exists.

A reviewer must inspect and reconstruct the small target region rather than treating the smooth surrounding ID3 prediction as an acceptable target contour. Small area alone does not identify a clinically small prostate or an apex/base slice. The tolerance counterexample above also explains why a permissive contour score would not be sufficient evidence of reliable small-region delineation.



Validation C: both models miss the nineteen-pixel reference region

Added boundary detail for case C, generated from the unchanged saved validation arrays; no new inference. The two models use the same display ROI, while all numbers use the full original image.



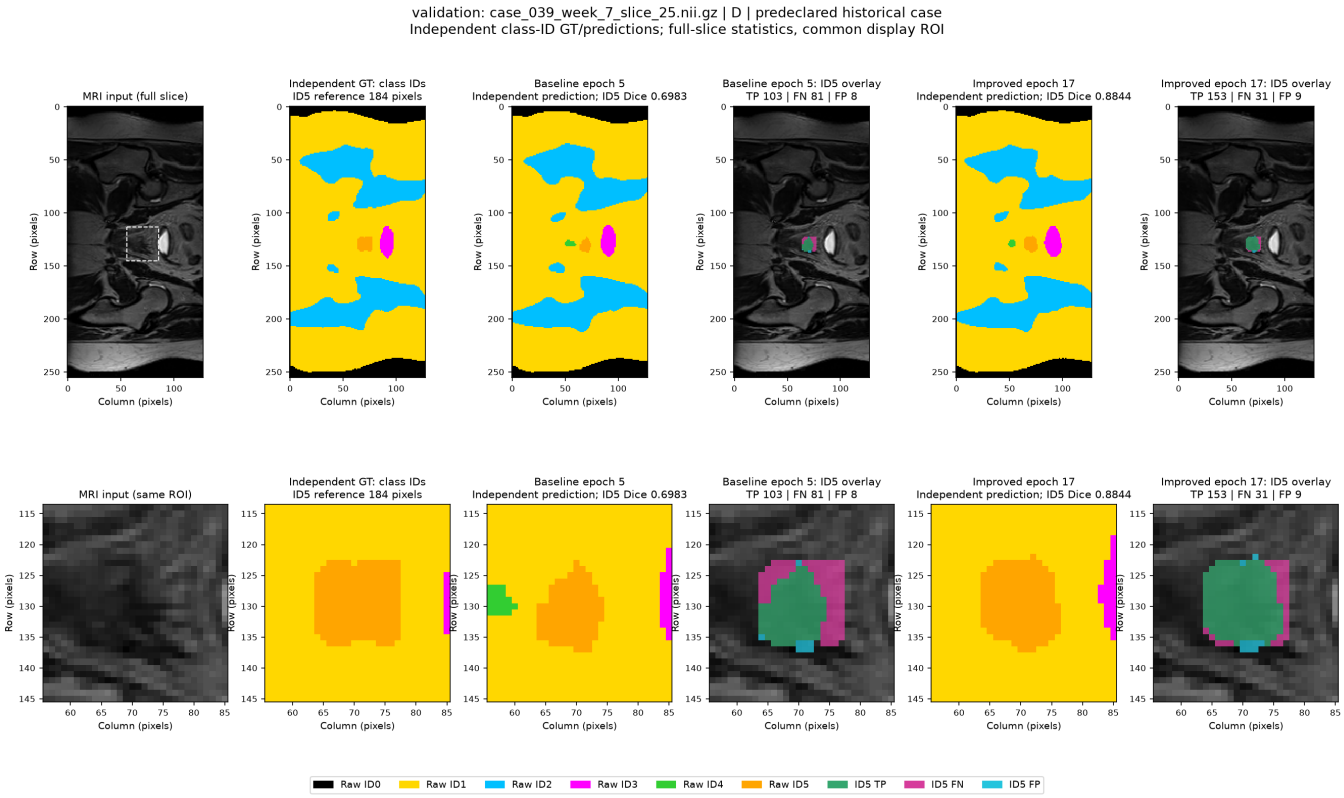
Added offline boundary detail for validation case C: reference and frozen-model contours from the same saved arrays

D. Improved overlap: case 039, week 7, slice 25

The reference ID5 region spans rows 123-136 and columns 64-77, and its annotated boundary adjoins ID1 (Body). Improved raises TP from 103 to 153 and reduces FN from 81 to 31 while FP changes from 8 to 9. A 26-pixel missed region remains on the larger-column side at rows 123-136, columns 74-77; the other five FN are at rows 134-136, columns 64-68. All 31 missed pixels are assigned to ID1.

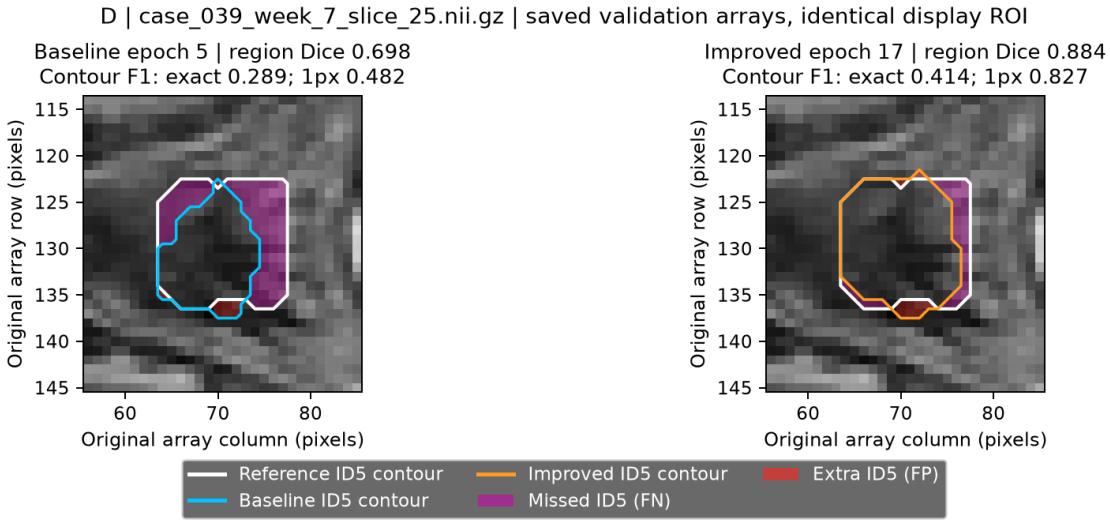
Improved also extends slightly beyond the reference: seven of its nine FP lie at rows 136-137, columns 70-73, with two further isolated FP pixels. These are local contour errors; the complete predicted ID5 region is still connected. Exact contour F1 improves from 0.289 to 0.414 and the 1px F1 from 0.482 to 0.827. A reviewer should restore the remaining larger-column edge and trim the small row-edge excess. The comparison supports local improvement on this chosen example, not a general

frequency or a verified anatomical-direction claim.



Validation D: increased true overlap and reduced missed region

Added boundary detail for case D, generated from the unchanged saved validation arrays; no new inference. The two models use the same display ROI, while all numbers use the full original image.



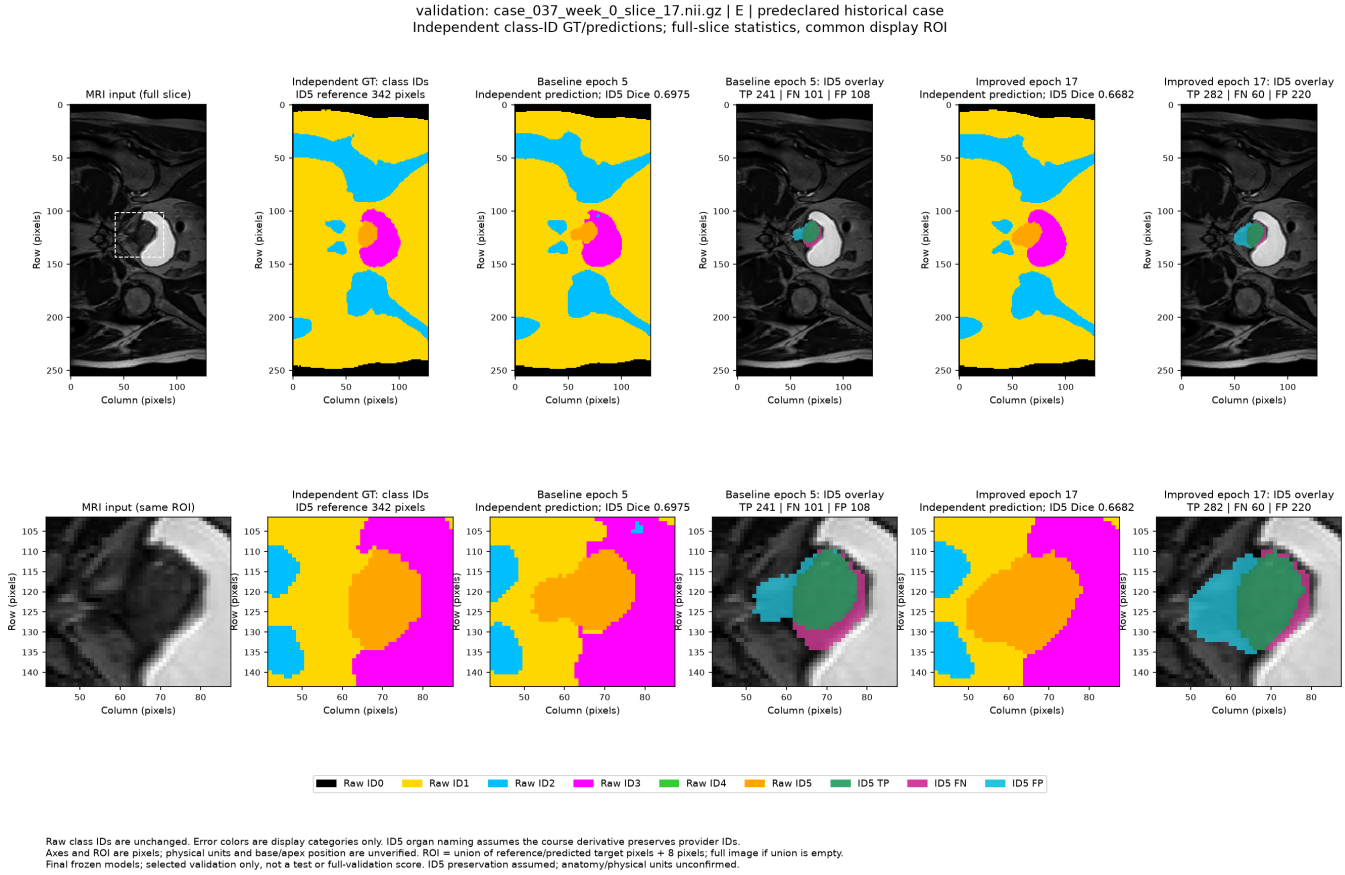
Added offline boundary detail for validation case D: reference and frozen-model contours from the same saved arrays

E. Over-segmentation cost: case 037, week 0, slice 17

The reference spans rows 110-134 and columns 62-79, with interfaces against reference IDs 1 and 3. Improved extends to column 50 although the reference starts at column 62. Its 220 FP are concentrated on the smaller-column side at rows 111-135, columns 50-66: 218 come from reference ID1 (Body) and two from ID3 (Bladder). On the other side, 60 reference ID5 pixels remain missed, all assigned to ID3. This combines an outward protrusion into the reference ID1 region with a residual loss of the ID5/ID3 interface.

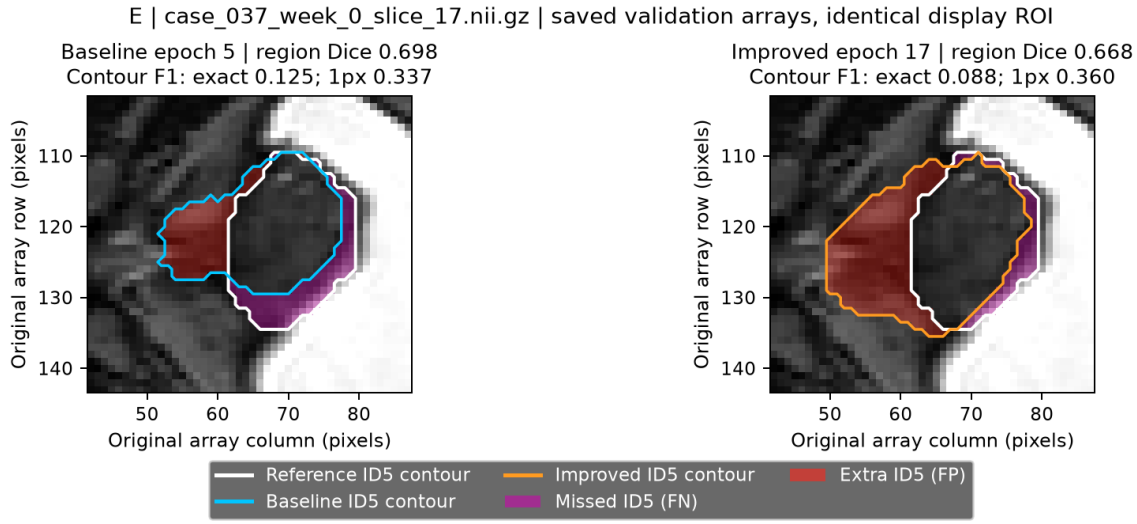
The metric conclusions are mixed: exact contour F1 falls from 0.125 to 0.088, while 1px F1 rises slightly from 0.337 to 0.360. Region Dice falls from 0.698 to 0.668 and FP more than doubles from 108 to 220. Selecting only the tolerant F1 would conceal

the larger over-segmentation cost. A reviewer should trim the smaller-column protrusion and inspect/redraw the still-missedID5/ID3 interface before accepting a draft; the increased TP alone is insufficient.



Validation E: reduced misses accompanied by substantial additional false positives

Added boundary detail for case E, generated from the unchanged saved validation arrays; no new inference. The two models use the same display ROI, while all numbers use the full original image.



Added offline boundary detail for validation case E: reference and frozen-model contours from the same saved arrays

These preselected examples are not random samples, failure rates, independent-patient statistics or a replacement for complete-split B4 scoring. In case E, the B1 baseline has 108 FP whereas the earlier B4 diagnostic has 109; checkpoint/input bytes were unchanged. Batch shape/composition and floating-point effects were not isolated, so no specific cause is claimed and the original B4 score is retained.

Coverage remains limited: every nonempty complete ID5 prediction in these five final-model cases has one four-connected and one eight-connected component. Small disconnected FP or FN error regions are not independent predicted organs. These selected examples therefore do not establish a disconnected-prediction failure mode. No physical units, independent patient

identity, anatomical base/apex position or adjacent-slice prediction continuity were verified. The contour diagnostics strengthenthis local pixel-space comparison but do not establish clinical boundary safety or resolve those unmeasured anatomicalquestions.

The earlier selection diagnosis and linked historical report preserve baseline-5/Improved-3/Improved-20 examples and their fixed selection rules. Those historical figures are neither final Improved-17 predictions nor test failures. The later final-model display evidence is separate from the original 19-source report summary.

7. Engineering recommendation and limitations

For a resource-sensitive prototype whose primary objective is ID5 overlap, the evidence does not establish enough benefit to recommend a default upgrade: primary Dice improves by only 0.106 percentage points, parameters increase 64.37%, recorded training allocated peak rises 3.11-fold, and measured synthetic forwards are slower. Retain the baseline as the low-cost engineering reference. This recommendation does not replace the pre-test selected Improved model or retrospectively select another checkpoint.

If reducing missed ID5 pixels or improving other classes is particularly valuable, Improved's higher recall and foreground macro scores may justify consideration, but extra FP and human-review costs must be counted. No predeclared clinical minimum meaningful difference, error-cost model or economic return is available. Neither statistical significance nor clinical benefit is claimed.

Evidence is limited to one seed, one derived 2D dataset and three test case-proxy groups. There is no independent patient crosswalk, no capacity-matched/module ablation, no equally weighted patient analysis, no clinical boundary review and no physical-distance metric. Prior implementation/test results had been encountered in earlier project work; this report does not claim that the data were universally unseen. The interruption and authorised repeated test exposure are disclosed above. Further tuning would require a new design and suitable independent data, not reuse of this sealed test set as a tuning set. Fragmented predictions, empty-reference FP, small missed regions and contour expansion require human inspection.

8. Files, dependencies and reproduction

Entry	Responsibility
modules.py	PyTorch model components and factory
dataset.py	Exact pairing, group audit, normalisation, padding and collate
train.py	Training/validation, curves and checkpoint roles
predict.py	Checkpoint restoration and single-slice displays; test examples require explicit opt-in
evaluate_frozen_test.py / recover_frozen_test.py	Separate shared-metric frozen testing and authorised recovery
README.md plus helper modules	Report and required losses, metrics, selection, resource and audit helpers

Submit the whole project directory rather than only the five named entry files. Raw data, checkpoints, environments and complete local run outputs are intentionally excluded from Git. Obtain data through the permitted course channel and prepare weights separately. An external data root must contain `keras_slices_train`, `keras_slices_seg_train`, `keras_slices_validate`, `keras_slices_seg_validate`, `keras_slices_test` and `keras_slices_seg_test`. Pass `--data-root` explicitly on another machine; defaults depend on the original computer layout.

requirements.txt pins direct dependencies; the observed CUDA build and 23-package runtime closure record the actual environment. Installation details and limitations are in DEPENDENCIES.md. A fresh-machine installation and cross-platform numerical reproduction have not been performed. PDF generation uses a separate document environment with ReportLab 4.4.9/Pillow 12.3.0, distinct from training-environment Pillow 13.0.0; see requirements-docs.txt.

From the repository root, this read-only archive command checks saved evidence, source bindings, syntax and links without importing model code or reading data/weights:

```
python recognition/hipmri_2d_improved_unet/verify_submission_archive.py
```

Run the command from the repository root with Git and the Python standard library. Original run records are retained; the archive verifier checks their exact source and evidence bindings. A complete clone is needed for historical-source verification; CUDA, model dependencies, data and checkpoints are not needed. To rebuild the report separately, install the document dependencies and choose a fresh PDF destination; existing files are not overwritten.

```
python recognition/hipmri_2d_improved_unet/export_readme_pdf.py --output outputs/readme_review_copy.pdf
```

The following are reproduction templates. Replace external paths and choose fresh output directories. Train a baseline by replacing `improved_unet` with `baseline_unet`. The ordinary training default selects CE; specify `id5_dice` for the final selection rule. Exact formal-plan/source bindings remain in the saved run plans; a new ordinary run is not the original frozen experiment.

```
python recognition/hipmri_2d_improved_unet/train.py --data-root 'C:\course-data\keras_slices_data' --output-dir 'outputs/new_t
raining_example' --model improved_unet --epochs 20 --batch-size 4 --base-channels 16 --num-classes 6 --lr 0.001 --seed 3710 --
device cuda --selection-metric id5_dice
python recognition/hipmri_2d_improved_unet/predict.py --data-root 'C:\course-data\keras_slices_data' --checkpoint 'C:\external
-checkpoints\model.pt' --split validation --filename case_038_week_0_slice_29.nii.gz --output-dir 'outputs/new_validation_exam
ple' --device cuda
```

Training saves summaries, curves, ID5-best, CE-best and last weights. Prediction saves `prediction.png`, `prediction_panels.png` and its summary. A B1 display does not replace the formal B4 score.

`predict.py --help` is safe to inspect without loading a framework, checkpoint or data. The test branch rejects execution before those imports/reads unless both `--split test` and `--allow-test-example` are supplied. That branch received 11 constructed CPU checks using temporary NIfTI/random tiny weights; **no real test example was generated**. A future real test-example forward requires a new explicit decision about test exposure. The constructed checks and source-evolution record distinguish the post-experiment display extension from original frozen code.

9. Evidence

The standard-library report collector reads 19 saved JSON sources and rechecks historical confusion matrices, complete scores and raw profile timings without reading MRI/masks/weights or running a framework. Its summary describes that earlier snapshot. The five later validation panels are separately bound by their display plan/record. Historic source hashes are not rewritten to describe newer documentation or display helpers.

Artificial Intelligence Usage Disclosure

Tools and extent. OpenAI Codex and its helper agents substantially assisted with task-sheet/paper interpretation, source checks, data/preprocessing and model code, the shared learning/evaluation pipeline, experiment execution, selection diagnostics, resource measurement, logging failure/recovery, evidence aggregation, report writing, diagrams, PDFs and Git operations. Code and automated experiment/verification commands were executed by Codex. This is not represented as independently implemented or personally run by the student.

Student participation. Recorded student contributions include learning responses about filename pairing, unchanged mask IDs, normalisation, valid versus padded pixels, tensor/batch dimensions, gradients and metrics; project/protocol and label-assumption decisions; and explicit training, test and Improved-only recovery authorisations. These responses and approvals are distinct from personally executing every check, auditing all data or demonstrating complete understanding of every submitted line.

Verification and limitations. Codex used constructed checks, smoke/full runs, checkpoint/state comparisons, saved-matrix/timing arithmetic, source hashes, image inspection and independent read-only reviews. Their scope and failures are preserved in the logs/JSON. Later work added 11 constructed CPU display checks, ten final-model validation forwards and 12 standard-library source-evolution checks. Automated validation does not prove student understanding or satisfy an originality judgement by itself. The historical implementation/test exposure, derivative-label assumption, logging failure and separately authorised repeated test exposure are disclosed rather than removed.

For the 27 October boundary supplement, Codex independently re-read only the five previously saved NPZ files, checked their recorded digests and existing matrices, computed the declared pixel-grid contour diagnostics, and drew additional contour comparisons. Fourteen constructed array-arithmetic checks support this custom diagnostic; they are not model or test-set evaluations. This supplement made zero raw NIfTI or checkpoint reads and zero model imports/forwards, training calls, test-set reads or optimiser updates. These automated checks and interpretations are not presented as independent student execution or as confirmation of anatomical base/apex labels.

Responsibility. The student remains responsible for reviewing, understanding and explaining the submitted code, formulae and claims, and for accurate attribution. AI assistance being permitted does not transfer that responsibility or guarantee a mark/pass.

References

- Ronneberger, O., Fischer, P., and Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. Baseline topology inspiration; this project's same padding, width and optimiser are implementation choices.

- Isensee, F., et al. (2018). Brain Tumor Segmentation and Radiomics Survival Prediction: Contribution to the BRATS 2017 Challenge. Architecture inspiration/course Ref. [2]; adapted to 2D rather than reproduced as the complete 3D recipe.
- CSIRO Data Access Portal. Prostate MRI collection, version 2, collection 52549. The source-label verification and course-derivative assumption are recorded in `label_schema.json`; this revision does not claim a fresh supporting-file retrieval.
- COMP3710 Report task sheet, 2026 v2.5, sections 2.2 and 3-8, PDF pages 5-6 and 12-17. Local assignment specification used for the project requirements and disclosure/checklist review.